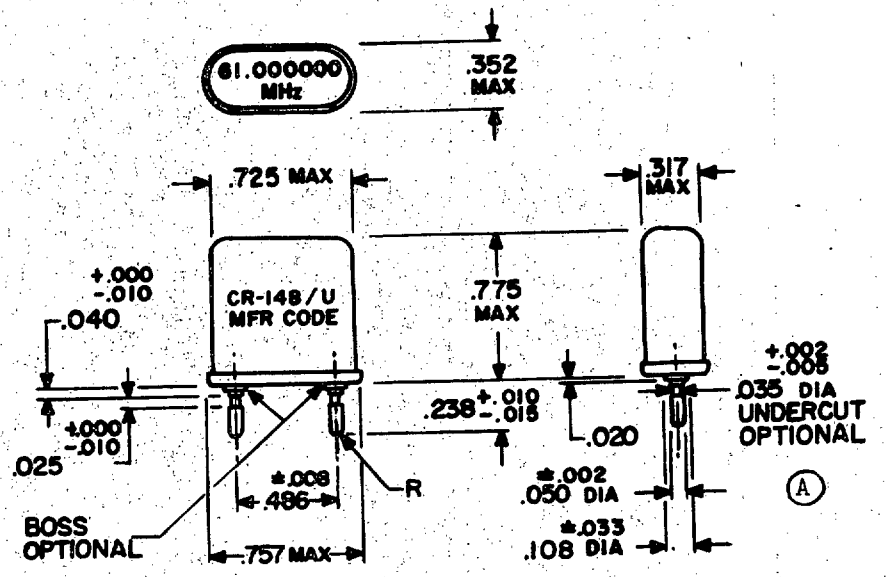
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-148/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit, quartz, described herein shall consist of this document and the latest issue of MIL-C-3098.

Pertinent characteristics: 10 to 61 MHz; third mechanical overtone; noncontrolled; series resonance.



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.020	.51	.050	1.27	.486	12.34
.005	.13	.025	.64	.108	2.74	.725	18.42
.008	.20	.033	.84	.238	6.05	.757	19.23
.010	.25	.035	.89	.317	8.05	.775	19.69
.015	.38	.040	1.02	.352	8.94		

NOTES:
1. DIMENSIONS ARE IN INCHES.
2. METRIC EQUIVALENTS ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 mm TO THE NEAREST .01 mm.

(A) Denotes Change

FSC 5955

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TABLE I. REQUIREMENTS

Requirement	Specified Value or Condition
Holder	HC-6/U (MIL-H-10056/2)
(A) Dimensions, Marking, and Configuration	See Figure
Frequency range	10.0 to 61.0 MHz, inclusive
Capacitance, shunt	7.0 pf, maximum
Mode of oscillation	Third mechanical overtone
Government reference standard test set	TS-683/TSM
Rated drive level	2.0 ±0.4, mw
Calibration values	
Equivalent resistance	51 ohms
Resistor voltage drop	.32 volt
Resonance	Series
Operating temperature range	-55° to +105°C, inclusive
Frequency tolerance	±0.005 percent
Equivalent resistance	40 ohms, maximum
Unwanted modes	Minimum unwanted-to-main mode resistance ratio of 2:1.
Shock:	
Frequency change permitted	±0.0005 percent
Equivalent resistance	±15%, or 2 ohms, whichever is greater
Vibration: 1/	
Frequency change permitted	±0.0005 percent
Equivalent resistance change permitted	±15%, or 2 ohms, whichever is greater
Temperature cycling:	
Frequency change permitted	±0.0005 percent
Equivalent resistance change permitted	±15%, or 2 ohms, whichever is greater
Aging:	
Frequency change permitted	0.0005%

1/ Method 201 of MIL-STD-202.

NOTE:

A. These crystal units were previously fabricated using Collins Radio Company's Specifications 289-7057-010/ -230, 290-9647/9656-00, and 291-7000/7009-00.

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Custodian:
Air Force - 84

Review activities:
Air Force - 11, 85

User activities:
Air Force - 17

Preparing activity:
Air Force - 84

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